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CViTLw: A Lightweight Convolutional Neural Network and Vision Transformer Hybrid Architecture for Plant Disease Classification

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ABSTRACT

Plant disease classification on edge devices requires models that are accurate, compact, and able to balance local visual cues with global context. Convolutional Neural Networks (CNNs) are effective for local lesion patterns, whereas Vision Transformers (ViTs) model long-range spatial relations but often require a larger parameter budget. This study proposes CViTLw, a lightweight parallel CNN–ViT architecture with 0.33 million parameters (0.33 M), consisting of MobileNetV2[:8], a 4-layer Micro-ViT branch, and a Cross-Fusion block with channel-spatial attention. The design is selected through a two-axis ablation study on attention placement and the role of the hybrid architecture. This study also introduces PlantRaw, a multi source field benchmark containing 16,165 images, 10 crop species, and 35 disease classes, to evaluate generalization under heterogeneous real-world images. CViTLw is evaluated on PlantPathology 2020, PlantVillage, PlantVillage Corn, Maize Leaf Disease, PlantRaw, and Indian Institute of Information Technology, Design and Manufacturing Jabalpur (IIITDMJ) Maize, with eight baselines including MobileNetV2, EfficientNet-B0, ResNet-50, ViT-B/16, Swin-Tiny, DeiT-Small, MobileViT-S, and ConvNeXt-Base. The model achieves accuracies of 97.16%, 99.73%, 98.44%, 99.44%, 90.94%, and 97.55%, respectively. Compared with the recent state-of-the-art (SOTA) ConViTX 2025 (0.70 M parameters), CViTLw reduces the parameter count by 0.37 M and obtains higher performance in most evaluated settings. These results indicate that CViTLw is a lightweight and parameter-efficient option for plant disease recognition under resource-constrained conditions.

1 | Introduction

Plant diseases remain one of the major causes of agricultural yield loss, with estimated reductions ranging from 20% to 40% depending on crop type and region [1]. This increases the need for automated disease detection in precision agriculture. A practical system must be accurate, compact, and efficient enough for deployment on resource-constrained edge devices such as unmanned aerial vehicles, mobile phones, or embedded monitoring units. Manual inspection by field experts is difficult to scale because it depends on expert availability, labor cost, and inspection time [2]. For this reason, computer vision and deep learning have become core tools for image-based plant disease recognition [2, 10, 32, 33].

Convolutional Neural Networks (CNNs) are effective for extracting local patterns such as lesions, lesion boundaries, and texture changes [10, 19, 20]. Lightweight network families such as MobileNetV2 (2018) [4] and EfficientNet [12] reduce the computational footprint while maintaining strong empirical performance. However, the local nature of convolution can limit the ability to model spatially diffuse symptoms or relations between distant leaf regions.

Vision Transformer (ViT) [5] and its hierarchical variants, such as Swin Transformer [6] and DeiT [30], represent global context more directly through self-attention. In plant disease tasks, this global modeling capability is practically meaningful because disease symptoms may spread across the leaf surface [22, 23, 24]. Nevertheless, many Transformer-based models still have large parameter counts and often require large-scale pretraining for stable optimization on domain-specific data, limiting direct use in resource-constrained agricultural scenarios.

Hybrid CNN–Transformer designs aim to combine the local feature extraction capability of CNNs with the global relation modeling capability of Transformers. Representative models such as MobileViT [13], CoAtNet [14], and LeViT [15] show that combining local convolutions with self-attention can produce more informative representations. In plant disease classification, recent CNN–ViT hybrid models, including the model by Thakur et al. [27] and ConViTX [28], demonstrate the potential of this direction in compact architectures. However, three gaps remain: (1) many hybrid models are still relatively heavy for deployment on resource-constrained edge devices; (2) design choices in hybrid architectures, such as the placement of attention blocks, are not always evaluated through systematic ablation; and (3) previous evaluations are often limited to one or two datasets, leaving the generalization capability of the proposed models insufficiently verified.

To address these gaps, this study proposes CViTLw, a lightweight CNN–Transformer hybrid architecture designed through a two-axis ablation study. From a representation learning perspective, the CNN branch captures high-frequency local features such as lesion morphology and texture, whereas the Transformer branch models low-frequency global structures such as disease distribution patterns and co-occurrence cues. The ablation study identifies an effective attention placement within the investigated design space, where Squeeze-and-Excitation (SE) and Convolutional Block Attention Module (CBAM) are applied only at the fusion stage, while also evaluating the role of the hybrid design against CNN-only and ViT-only alternatives. The resulting model is then validated using two evaluation layers: (1) standardized comparison with eight baselines, namely MobileNetV2, EfficientNet-B0, ResNet-50, ViT-B/16, Swin-Tiny, DeiT-Small, MobileViT-S, and ConvNeXt-Base, across six datasets under unified protocols; and (2) comparison with published state-of-the-art (SOTA) results on PlantVillage and IIITDMJ Maize, including ConViTX 2025, to clarify the novelty and parameter efficiency of the proposed model.

The main contributions of this study are as follows:

1. We propose CViTLw, a lightweight and parameter-efficient CNN–Vision Transformer hybrid architecture with approximately 0.33 million parameters (0.33 M), combining a MobileNetV2[:8] branch, a 4-layer Micro-ViT branch, and an SE+CBAM Cross-Fusion block to merge local and global features.
2. We establish the design basis through a two-axis ablation study covering attention placement and the role of the hybrid architecture compared with CNN-only and ViT-only variants.
3. We conduct a controlled comparison with eight representative baselines, namely MobileNetV2, EfficientNet-B0, ResNet-50, ViT-B/16, Swin-Tiny, DeiT-Small, MobileViT-S, and ConvNeXt-Base, and further compare CViTLw with recent SOTA studies on PlantVillage and IIITDMJ Maize, including ConViTX 2025.
4. We introduce PlantRaw [35], a multi source field benchmark containing 16,165 images, 10 crop species, and 35 disease classes. Across six datasets, CViTLw achieves 97.16% on Plant-Pathology 2020, 99.73% on PlantVillage, 98.44% on PlantVillage Corn, 99.44% on Maize Leaf Disease, 90.94% on PlantRaw, and 97.55% in transfer learning on IIITDMJ Maize.

To foster reproducible research and facilitate model verification, the complete source code, model inspection notebook, raw training logs, JSON metrics, and trained model weights for all six datasets are publicly released at GitHub: https://github.com/tonglethanhhai/CViTLw.

The remainder of this paper is organized as follows. Section 2 reviews related studies. Section 3 describes the proposed method, architectural design, and ablation rationale. Section 4 presents the experimental setup. Section 5 analyzes quantitative and qualitative results. Section 6 discusses interpretability, generalization, and study scope. Section 7 concludes the paper.

2 | Related Work

2.1 | CNN-Based Plant Disease Classification

Lightweight CNNs, including MobileNet [9], MobileNetV2 (2018) [4], and EfficientNet [12], have become common choices for plant disease recognition because of their favorable balance between accuracy and FLOPs [2, 10]. Attention modules such as SE and CBAM can improve performance by emphasizing image regions with important diagnostic information [19, 20, 21, 25, 26]. More recently, ConvNeXt [31] has shown that convolution-based architectures can be competitive with Transformers when using modern Transformer training practices. However, CNNs with a single branch remain constrained by local receptive fields, limiting their ability to capture disease patterns distributed across the whole leaf.

2.2 | Vision Transformers for Image Classification

ViT [5] and hierarchical variants such as Swin Transformer [6] and DeiT [30] model patch-wise image relations through self-attention, which is useful for plant disease problems where pathological cues may extend over large spatial areas [22, 23, 24]. The trade-off is the higher parameter count and dependence on large-scale pretraining. DeiT improves data efficiency through knowledge distillation, whereas Swin reduces computational cost through shifted-window attention.

2.3 | Hybrid CNN–ViT Architectures

Hybrid CNN–Transformer architectures are designed to combine the local feature encoding advantage of convolution with the long-range dependency modeling capability of self-attention. LeViT [15], MobileViT [13], and CoAtNet [14] are representative designs in general computer vision. For plant disease classification and compact models, this study uses MobileViT-S [13] as a hybrid baseline in the quantitative experiments and ConViTX [28]

TABLE 1 | Architectural comparison between CViTLw and closely related CNN–Transformer hybrid models.

Approach	Params.	Custom ViT	Attention	Multi-dataset Evaluation
MobileViT-S (2022) [13]	5.60 M	No	SE	No
ConViTX (2025) [28]	0.70 M	Yes	SE	Yes (2)
CViTLw (Ours)	**0.33 M**	**Yes**	**SE+CBAM**	**Yes (6)**

as a recent SOTA reference for contextual comparison. Table 1 summarizes the design differences between CViTLw and the hybrid architectures most closely related to this study, focusing on parameter count, custom ViT design, attention mechanism, and multi-dataset evaluation scope.

In plant disease studies, ConViTX by Thakur et al. [28] is a parallel CNN–ViT hybrid model evaluated on field maize data and PlantVillage. Compared with this approach, CViTLw preserves the parallel CNN–ViT design principle but reduces the parameter count, replaces the single SE mechanism with SE+CBAM at the fusion stage, and extends the evaluation scope to six datasets, including the multi source PlantRaw benchmark. In addition, this study adds a two-axis ablation analysis to investigate attention placement and the role of the hybrid structure, thereby clarifying the design basis of the proposed model.

2.4 | Attention Mechanisms

SE [7] performs channel-wise recalibration, whereas CBAM [8] adds a spatial branch. Although these mechanisms are widely used in plant disease recognition [19, 20, 21, 25, 26], the *placement* of attention in a hybrid architecture, including branch-level, fusion-level, or multi-stage placement, still requires systematic investigation. The ablation study in Section 5.1 addresses this gap using nine configurations and reports the corresponding statistical results.

3 | Proposed Method

3.1 | CViTLw Architecture and Design Rationale

Fig. 1 illustrates the complete processing flow of CViTLw. An input red-green-blue (RGB) image of size $224 \times 224 \times 3$ is simultaneously fed into two parallel branches: (i) a *local CNN branch* (MobileNetV2, the first 8 inverted residual blocks), which extracts high-frequency local features such as lesion morphology and texture gradients; and (ii) a *global Transformer branch* (custom lightweight ViT, 4 encoder layers), which captures low-frequency structural patterns including disease distribution over the leaf surface and inter-region correlations. The two feature streams are merged in the cross-fusion block through channel-wise concatenation, processed by an inverted residual block, refined by SE+CBAM attention, and classified using Global Average Pooling (GAP) and a fully connected (FC) layer.

Table 2 provides a system-level architectural summary that serves as a quick specification for following the detailed descriptions in Sections III-B–III-F. This presentation clarifies the tensor transformation sequence, parameter budget, and design role of each component before the module-level analysis.

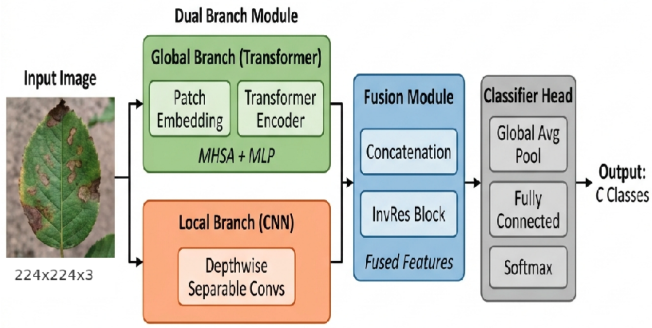

FIGURE 1 | Overview of the CViTLw Processing Pipeline.

3.2 | Lightweight CNN Branch (MobileNetV2[:8])

The CNN branch uses the first eight inverted residual blocks of MobileNetV2 (2018) [4]. Each inverted residual block applies depthwise separable convolution with an expansion factor t:

$$\mathbf{y} = \mathbf{x} + \mathrm{PW}_{\downarrow}(\mathrm{DW}(\mathrm{PW}_{\uparrow}(\mathbf{x}))) \tag{1}$$

where PW denotes pointwise convolution, DW denotes depthwise convolution, $\mathrm{PW}_{\uparrow}$ expands the number of channels by factor t, and $\mathrm{PW}_{\downarrow}$ projects them back. The residual shortcut is added only when the input and output dimensions match [4].

Table 3 gives the layer-wise specification of the CNN branch. The eight MobileNetV2 blocks with width multiplier 1.0 produce a $14\times14\times64$ feature map, which is projected to 32 channels by a 1×1 convolution followed by batch normalization (BN) and Rectified Linear Unit (ReLU). Before fusion, this map is bilinearly resized to $16 \times 16 \times 32$ so that it is spatially aligned with the Micro-ViT output.

$$F_{\mathrm{CNN}} = \mathrm{ReLU}\Big(\mathrm{BN}\Big(\mathrm{Conv}_{1\times1}^{64\to32}\left(\mathrm{MobileNetV2}_{[:8]}(\mathbf{x})\right)\Big)\Big) \in \mathbb{R}^{14\times14\times32} \tag{2}$$

Rationale for selecting the first eight blocks. Using only the first eight blocks preserves a moderate spatial resolution (14×14) while extracting sufficiently rich local features. The remaining MobileNetV2 blocks mainly increase channel depth without adding spatial resolution, which would increase the parameter count without providing a clear benefit for the compact fusion design considered in this study.

Parameter count of the CNN branch. With width multiplier 1.0, MobileNetV2 blocks 0–7 contain approximately 0.08 M parameters. The 1×1 projection adds $64 \times 32 + 32 \approx 0.002$ M. The low parameter count is mainly due to the depthwise separable convolution design of MobileNetV2, which factorizes standard convolution into depthwise and pointwise operations, substantially reducing parameters and FLOPs.

3.3 | Micro-Vision Transformer Branch

The Transformer branch of CViTLw captures long-range spatial dependencies that local convolution cannot model efficiently. We follow the formulation of Dosovitskiy et al. [5] and the exposition in Zhang et al. [18].

TABLE 2 | Architectural specification summary of CViTLw.

Stage	Main operation	Output tensor	Parameters	Design role
Input	Resize and normalize RGB leaf image	224×224×3	–	Standardized input resolution for all datasets.
CNN branch	MobileNetV2[:8] + 1×1 projection	14×14×32	0.082 M	Local lesion texture, edge, and color-pattern features with strong convolutional inductive bias.
Micro-ViT branch	7×7 patch embedding + 4-layer Transformer + depthwise convolution (DWConv) smoothing	16×16×48	0.137 M	Long-range spatial context and global disease distribution over the leaf surface.
Cross-fusion	Resize CNN map (14×14×32 → 16×16×32), concatenate with ViT map (16×16×48), InvRes(80→128)	16×16×80 → 8×8×128	0.070 M	Joint local and global representation in a compact feature space.
Attention refinement	Sequential SE + CBAM applied only after fusion	8×8×128	0.010 M	Channel and spatial reweighting of fused features with low overhead.
Classifier	GAP + FC(128→C)	C classes	≤0.005 M	Dataset-specific prediction head.
Total	Overall lightweight hybrid CNN–ViT model	–	**0.329–0.334 M**	Depends on the number of classes C.

TABLE 3 | Feature-size flow of the CNN Branch before cross-fusion. InvRes(c_{out}, t, s) denotes an inverted residual block with output channels c_{out}, expansion factor t, and stride s.

Block	Operator	Output Channels	Stride	Output Shape
0	Conv2d(3×3) + BN + ReLU6	32	2	$112 \times 112 \times 32$
1	InvRes(16, $t = 1, s = 1$)	16	1	$112 \times 112 \times 16$
2	InvRes(24, $t = 6, s = 2$) ×2	24	2	$56 \times 56 \times 24$
3	InvRes(32, $t = 6, s = 2$) ×3	32	2	$28 \times 28 \times 32$
4	InvRes(64, $t = 6, s = 2$) ×4	64	2	$14 \times 14 \times 64$
–	Conv2d(1×1) + BN + ReLU	32	1	$14 \times 14 \times 32$
–	Bilinear resize for fusion	32	–	$\mathbf{16 \times 16 \times 32}$

TABLE 4 | Feature-size flow of the Micro-ViT Branch before cross-fusion. All DWConv layers use 3×3 kernels with groups = 48.

Stage	Operator	Output Shape
Patch embed	Conv2d(k=7, s=7, 3→48)	$32 \times 32 \times 48$
Pos. encode	Add $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{1024\times48}$	$B \times 1024 \times 48$
Transformer	4× [MHA(h=4, d_k=12) + FFN(d_{ff}=96)]	$B \times 1024 \times 48$
LayerNorm	LN + reshape to 2D	$32 \times 32 \times 48$
DWConv-1	DWConv(k=3, s=1, p=1) + BN + ReLU	$32 \times 32 \times 48$
DWConv-2	DWConv(k=3, s=1, p=1) + BN + ReLU	$32 \times 32 \times 48$
DWConv-3	DWConv(k=3, s=2, p=1) + BN + ReLU	$\mathbf{16 \times 16 \times 48}$
Fusion input	Feature map used for channel-wise concatenation	$\mathbf{16 \times 16 \times 48}$

3.3.1 | Patch Embedding

An input image $\mathbf{x} \in \mathbb{R}^{224\times224\times3}$ is split into non-overlapping patches of size $P \times P = 7 \times 7$ using a convolutional projection:

$$\mathbf{Z}_0 = \text{Conv2d}^{3\to48}_{k=7,s=7}(\mathbf{x}) \in \mathbb{R}^{32\times32\times48} \tag{3}$$

This operation produces $N = (224/7)^2 = 1024$ patch tokens, each with embedding dimension $d = 48$.

3.3.2 | Positional Encoding

A learnable positional encoding $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{1\times1024\times48}$ is added to preserve spatial order information:

$$\mathbf{Z}'_0 = \text{flatten}(\mathbf{Z}_0) + \mathbf{E}_{\text{pos}} \in \mathbb{R}^{B\times1024\times48} \tag{4}$$

3.3.3 | Transformer Encoder

The token sequence passes through $L = 4$ standard Transformer encoder layers. Each layer consists of Multi-Head Self-Attention (MHA) [16] and a Feed-Forward Network (FFN), with pre-norm Layer Normalization (LN) [17] and residual connections:

$$\mathbf{z}'_\ell = \text{MHA}(\text{LN}(\mathbf{z}_{\ell-1})) + \mathbf{z}_{\ell-1} \tag{5}$$

$$\mathbf{z}_\ell = \text{FFN}(\text{LN}(\mathbf{z}'_\ell)) + \mathbf{z}'_\ell \tag{6}$$

Multi-head self-attention uses four heads ($h = 4$), with $d_k = d/h = 12$:

$$\text{MHA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\text{head}_1, \dots, \text{head}_4)\,\mathbf{W}^O \tag{7}$$

$$\text{head}_i = \text{softmax}\left(\frac{\mathbf{QW}_i^Q(\mathbf{KW}_i^K)^\top}{\sqrt{d_k}}\right)\mathbf{VW}_i^V \tag{8}$$

The FFN uses the Gaussian Error Linear Unit (GELU) activation with hidden dimension $d_{ff} = 2d = 96$:

$$\text{FFN}(\mathbf{x}) = \text{GELU}(\mathbf{xW}_1 + \mathbf{b}_1)\mathbf{W}_2 + \mathbf{b}_2 \tag{9}$$

3.3.4 | Spatial Smoothing via Depthwise Convolution

After $L = 4$ encoder layers, a final LayerNorm is applied and the token sequence is reshaped back to a 2D feature map of size $32 \times 32 \times 48$. Three depthwise convolution layers progressively smooth the spatial representation:

$$\begin{aligned} \tilde{F} &= \text{reshape}(\text{LN}(\mathbf{z}_L)), \\ F_{\text{ViT}} &= \text{DWConv}_{s=2}(\text{DWConv}_{s=1}(\text{DWConv}_{s=1}(\tilde{F}))) \\ &\in \mathbb{R}^{16\times16\times48} \end{aligned} \tag{10}$$

Table 4 gives the layer-wise specification. After the final DWConv layer, the Micro-ViT branch produces a $16 \times 16 \times 48$ feature map, which has the same spatial size as the resized CNN feature map before channel concatenation.

The three depthwise convolutions serve two purposes: (1) the final stride-2 convolution reduces the 32×32 grid to 16×16, matching the resized spatial resolution of the CNN branch; and (2) the two stride-1 convolutions provide local spatial smoothing, bridging the gap between token-level global attention and the convolutional fusion block.

Parameter count of the ViT branch: Each Transformer layer contains $4d^2$ (MHA) $+2d \cdot d_{ff}$ (FFN) $\approx 4\times48^2 + 2\times48\times96 = 9,216+$

$9,216 = 18,432$ parameters, excluding bias and LN. With four layers, this gives $\approx 73,728$ parameters. Adding patch embedding ($3\times7\times7\times48 = 7,056$), positional encoding ($1,024\times48 = 49,152$), DWConv layers, and normalization terms gives approximately 0.13–0.14 M parameters.

3.4 | Cross-Fusion Re-Attention Block (SE + CBAM)

The cross-fusion block projects local and global signals into a shared representation space. Before concatenation, the CNN branch produces a $14 \times 14 \times 32$ feature map, whereas the ViT branch outputs a $16 \times 16 \times 48$ feature map. To align the two branches, the CNN feature map is bilinearly interpolated to $16 \times 16 \times 32$:

$$F_{\text{fused}} = [\text{Resize}_{14\times14\to16\times16}(F_{\text{CNN}});\ F_{\text{ViT}}] \in \mathbb{R}^{16\times16\times(32+48)} = \mathbb{R}^{16\times16\times80} \tag{11}$$

The fused tensor is processed by an inverted residual block to project the concatenated 80-channel sequence to 128 channels while halving the spatial resolution:

$$F_{\text{mixed}} = \text{InvRes}_{80\to128}^{t=6,s=2}(F_{\text{fused}}) \in \mathbb{R}^{8\times8\times128} \tag{12}$$

The SE+CBAM attention module (Section 3.5) then refines F_{mixed}:

$$\hat{y} = \text{FC}_{128\to C}(\text{GAP}(\text{CBAM}(\text{SE}(F_{\text{mixed}})))) \in \mathbb{R}^{C} \tag{13}$$

where C is the number of output classes. The resize step maps the CNN branch to the original Transformer spatial grid instead of forcing the Transformer to match the CNN grid, preserving the positional encoding structure of the Transformer.

Parameter count of the fusion block: The InvRes block ($80 \to 128$, t=6) expands $80\times6 = 480$ channels, applies a 3×3 depthwise convolution, and projects to 128 channels, requiring approximately 0.07 M parameters. SE (r=4) requires $2 \times 128^2/4 = 8,192$ parameters. Channel CBAM (r=16) requires $2 \times 128^2/16 = 2,048$ parameters, and spatial CBAM with a 7×7 convolution requires 98 parameters. The FC head ($128 \times C$) depends on C. The fusion block and head together require approximately 0.08–0.09 M parameters.

3.5 | Attention Blocks

3.5.1 | SE (Squeeze-and-Excitation)

The SE block [7] performs channel-wise recalibration:

$$s = \sigma(W_2\,\delta(W_1\,\text{GAP}(F))) \tag{14}$$

$$F_{\text{out}} = s \otimes F \tag{15}$$

where δ denotes ReLU, σ denotes sigmoid, and the reduction ratio is $r = 4$.

3.5.2 | CBAM (Convolutional Block Attention Module)

CBAM [8] extends SE by adding spatial attention:

Channel attention:

$$M_c(F) = \sigma(\text{FC}(\delta(\text{FC}(\text{GAP}(F)))) + \text{FC}(\delta(\text{FC}(\text{MaxPool}(F))))) \tag{16}$$

$$F' = M_c(F) \otimes F \tag{17}$$

Spatial attention:

$$M_s(F') = \sigma(\text{Conv}_{7\times7}([\text{AvgPool}_{ch}(F');\ \text{MaxPool}_{ch}(F')])) \tag{18}$$

$$F_{\text{out}} = M_s(F') \otimes F' \tag{19}$$

In CViTLw, SE and CBAM are applied *sequentially* only at the fusion stage, namely SE followed by CBAM on the fused 128-channel features. This placement choice is empirically evaluated through the ablation study in Section 5.1.

TABLE 5 | Detailed module-wise parameter breakdown of CViTLw.

Module	Parameters	Ratio (%)
CNN Branch (MobileNetV2[:8] + 1×1 proj.)	0.079 M (78,656)	23.6%
Micro-ViT Branch (Patch Embed. + 4L Encoder + DWConv)	0.134 M (133,776)	40.1%
InvRes Fusion Block (Concat 80→128)	0.106 M (106,336)	31.9%
SE + CBAM Attention Module	0.010 M (10,338)	3.1%
Classifier Head FC (128→ C)	0.0004–0.005 M	0.1–1.3%
Total (C=38, PlantVillage Full [11])	**0.334 M**	100%
Total (C=35, PlantRaw (Ours), 2026)	**0.334 M**	100%
Total (C=4, PlantPathology 2020 [34])	**0.330 M**	100%
Total (C=4, PV-Corn Subset [11])	**0.330 M**	100%
Total (C=4, IIITDMJ Maize [28])	**0.330 M**	100%
Total (C=3, Maize Leaf Disease [29])	**0.329 M**	100%

3.6 | Parameter and Complexity Analysis

Table 5 reports the verified parameter count for each module. The total is 0.329–0.334 M, depending on the number of output classes.

The parameter distribution across the CNN branch (23.6%), Micro-ViT branch (40.1%), fusion block (31.9%), and attention module (3.1%) shows that the model budget is allocated to local feature extraction, global context modeling, and feature fusion.

Rationale for controlling the parameter budget. Three design choices keep the parameter count below 0.35 M: (1) the CNN branch uses only the first 8 MobileNetV2 blocks with depthwise separable convolution instead of the full network; (2) the custom ViT uses $d = 48$ instead of 768 in ViT-B/16, $h = 4$ heads instead of 12, and $L = 4$ layers instead of 12; and (3) the DWConv smoothing layers use grouped operations and therefore add only a small number of parameters.

3.7 | Architectural Data Flow and Algorithm

This subsection presents CViTLw at two complementary description levels. Fig. 2 illustrates the data flow and tensor dimensions at each stage, whereas Algorithm 1 summarizes the processing sequence from the input image to output prediction and parameter update during training.

In the detailed data flow, the ViT branch transforms the input from $224\times224\times3$ to $32\times32\times48$ and then to $16\times16\times48$. The CNN branch produces features through the sequence $224 \times 224 \times 3 \to 14 \times 14 \times 64 \to 14 \times 14 \times 32 \to 16 \times 16 \times 32$. The two branches are concatenated into a $16 \times 16 \times 80$ tensor, compressed and refined to $8 \times 8 \times 128$, and finally mapped to C output classes.

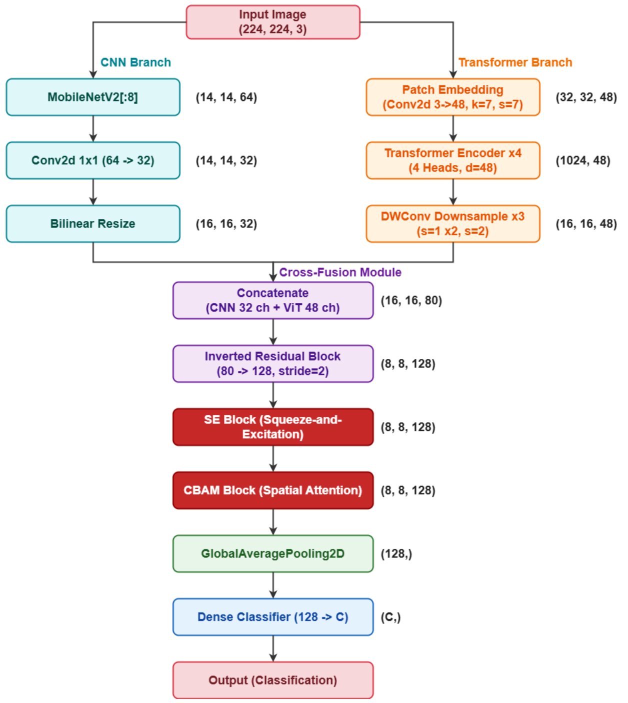

FIGURE 2 | Detailed Data Flow of CViTLw.

Algorithm 1: CViTLw Training and Inference Procedure

Input: RGB leaf image I, ground-truth label y during training, number of classes C
Output: Class-probability vector $\hat{\mathbf{y}} \in \mathbb{R}^{C}$
Convention: Algorithmic tensors use batch-channel-height-width format; architectural equations omit the batch dimension and use height-width-channel format.

1. Resize and normalize I to obtain $I_{\text{norm}} \in \mathbb{R}^{B\times 3\times 224\times 224}$.
2. Extract local features with the first eight MobileNetV2 inverted residual blocks and a 1×1 projection: $F_{\text{cnn}} \in \mathbb{R}^{B\times 32\times 14\times 14}$.
3. Extract global-context features with a 7×7 patch embedding, four Micro-ViT encoder blocks, reshaping, and three DWConv layers: $F_{\text{vit}} \in \mathbb{R}^{B\times 48\times 16\times 16}$.
4. Resize F_{cnn} to 16×16 and concatenate it with F_{vit} along the channel dimension: $F_{\text{fused}} \in \mathbb{R}^{B\times 80\times 16\times 16}$.
5. Project the fused tensor with an inverted residual fusion block and refine it using sequential SE and CBAM attention: $F_{\text{att}} \in \mathbb{R}^{B\times 128\times 8\times 8}$.
6. Apply global average pooling and a fully connected classifier to obtain logits $\mathbf{o} = \text{FC}(\text{GAP}(F_{\text{att}}))$.
7. Compute class probabilities $\hat{\mathbf{y}} = \text{softmax}(\mathbf{o})$.
8. If training, compute cross-entropy loss $\mathcal{L}(\hat{\mathbf{y}}, y)$ and update model parameters with Adam.

Algorithm 1 expresses the same procedure as key operational steps, including image preprocessing, parallel feature extraction, feature fusion, loss computation, and parameter update. This presentation clarifies the relationship between the architectural diagram and the training/inference procedure used in the experiments.

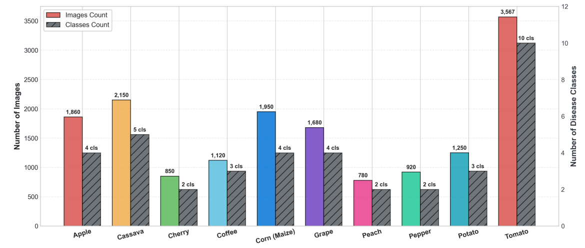

FIGURE 3 | PlantRaw Dataset Composition.

4 | Experimental Setup

4.1 | Datasets and Acquisition Settings

To evaluate CViTLw under diverse data conditions, this study selects six experimental datasets that differ in acquisition environment, sample size, number of classes, crop species, and domain-transfer scenario. Table 6 summarizes the six datasets used in this study.

Scientific rationale for selecting the six datasets: The six datasets were selected to evaluate CViTLw under common image-based plant disease recognition conditions, including the domain gap between lab and field images, variation in sample scale, differences in class count, and viewpoint changes in transfer learning. Therefore, this data setup not only measures accuracy under controlled conditions but also examines generalization under more realistic and heterogeneous settings.

1. Acquisition-condition coverage (controlled laboratory vs. in-situ field): The combination of standardized laboratory images (PlantVillage [11]) and complex field images (Maize Leaf Disease [29], PlantPathology 2020 [34]) evaluates the model under both controlled and real-world imaging conditions.
2. Single-crop and multi-crop evaluation at different scales: Single-crop datasets (Maize [29], Apple [34]) allow detailed assessment of narrow disease symptom morphologies, whereas large multi-crop datasets (38-class PlantVillage [11] and 35-class PlantRaw) evaluate scalability across crop species and disease categories.
3. PlantRaw field data challenge: PlantRaw integrates images from nine heterogeneous open repositories covering 10 major crop species, evaluating model generalization under real-world noise factors such as different acquisition devices, variable resolutions, and uncurated images. In particular, PlantRaw has a high class imbalance ratio, which is a common challenge in machine learning because models may favor majority classes.
4. Cross-domain transfer and drone viewpoint evaluation (IIIT-DMJ Maize [28]): This dataset tests transfer learning from laboratory knowledge to a small field dataset and evaluates robustness to aerial drone viewpoints that differ from standard leaf close-up images.

Fig. 3 illustrates the PlantRaw composition by crop group and disease class, providing visual context for the multi source field benchmark used in this study.

TABLE 6 | Overview of the six evaluation datasets used in this study.

Dataset	Year	Samples	Plants	Classes	Imb. Ratio	Acquisition Setup / Location	Primary Evaluation Purpose
PlantVillage Full [11]	2016	54,305	14 Crops	38	36.2×	Controlled Laboratory	Standard Lab Benchmark & Pretraining
PV-Corn Subset [11]	2016	3,852	Maize	4	2.3×	Controlled Laboratory	Lab-controlled Corn Benchmark
PlantPathology 2020 [34]	2020	6,192	Apple	4	12.2×	Real-world field (Canon DSLR)	2-Axis Ablation Study
Maize Leaf Disease [29]	2023	18,402	Maize	3	2.3×	Real-world field (Smartphones)	In-situ Field Evaluation
IIITDMJ Maize [28]	2025	416 / 816 / 200	Maize	4	1.2×	Real-world Field & Aerial Drone	Cross-domain Transfer Learning
PlantRaw (Ours) [35]	2026	16,165	10 Crops	35	164.5×	Real-world multi-source (Internet)	Wild Multi-crop Generalization

Note: Imb. Ratio = max class size / min class size. Sample count for IIITDMJ Maize indicates 416 raw field images, 816 augmented images, and 200 unseen aerial drone images for out-of-domain evaluation.

TABLE 7 | Experimental Protocol and Implementation Environment.

Component	Setting
Training and validation protocol	
Loss function	Cross-entropy.
Optimizer	Adam with initial learning rate 10^{-4}.
Learning-rate schedule	CosineAnnealingLR with $T_{\max} = 50$, decaying from 10^{-4} to 10^{-6}.
Batch size and epochs	Batch size is model-dependent: 64 for CViTLw and up to 256 for lightweight baselines; the maximum number of epochs is 50.
Early stopping	Patience of 15 epochs monitored on validation loss.
Validation protocol	Stratified 5-fold cross-validation with seed 42 for all datasets, except PlantRaw, which follows the predefined 80/20 split.
Preprocessing and augmentation	
Image preprocessing	All images are resized to $224 \times 224 \times 3$ and normalized with $\mu = [0.485; 0.456; 0.406]$ and $\sigma = [0.229; 0.224; 0.225]$.
Data augmentation	Random horizontal flip ($p = 0.5$), random rotation ($\pm 20°$), and color jitter with brightness ± 0.2 and contrast ± 0.2.
Model initialization	For the primary comparisons, CViTLw and all baseline models are trained from scratch with random initialization and without pretrained weights. The only exception is the explicitly defined transfer-learning experiment on IIITDMJ Maize, where PlantVillage-pretrained weights are used. This setting reduces the influence of external pretraining and isolates the role of architecture.
Implementation environment	
Hardware platform	Intel Core i5-13400F CPU, 32 GB DDR5 RAM, NVIDIA GeForce RTX 5070 Ti GPU with 16 GB VRAM, and 500 GB NVMe SSD.
Operating system	Ubuntu 24.04 LTS, 64-bit.
Software stack	Python 3.12.10; PyTorch with CUDA acceleration; torchvision, Pillow, NumPy, pandas, scikit-learn, and thop 0.1.1.

4.2 | Training Protocol and Environment

Table 7 summarizes the training protocol, data preprocessing, validation strategy, and experimental environment used in this study. CViTLw and the comparison models are evaluated under the same training setup to reduce differences caused by optimization procedures and to better isolate the influence of model architecture.

4.3 | Comparison Baselines and SOTA References

Eight representative architectures are selected as standardized baselines, covering three main design groups: CNN-based models, Transformer-based models, and CNN–Transformer hybrid models:

- CNN-based models: MobileNetV2 (2018) [4] (2.23 M), EfficientNet-B0 (2019) [12] (4.01 M), ResNet-50 (2016) [3] (25.56 M), ConvNeXt-Base (2022) [31] (87.57 M).
- Transformer-based models: ViT-B/16 (2021) [5] (85.80 M), Swin-Tiny (2021) [6] (27.52 M), DeiT-Small (2021) [30] (22.05 M).
- CNN–Transformer hybrid model: MobileViT-S (2022) [13] (5.60 M).

In addition to baseline comparisons under a unified protocol, this study compares CViTLw with published state-of-the-art (SOTA) results on two datasets with direct comparison data: PlantVillage and IIITDMJ Maize. On PlantVillage, CViTLw is compared with published studies on the same benchmark, including ConViTX (2025) [28]. On IIITDMJ Maize, the comparison focuses on ConViTX (2025) and other published methods reported for the same dataset. These comparisons are reported separately from the controlled baseline tables because published studies may differ in training protocol, data splitting, preprocessing, and implementation details.

4.4 | Evaluation Metrics

This study uses standard classification evaluation metrics [18]. In the following equations, TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively:

$$\text{Accuracy (AC)} = \frac{\text{TP} + \text{TN}}{\text{Total}} \tag{20}$$

$$\text{Precision (PR)} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{21}$$

$$\text{Recall (RE)} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{22}$$

$$\text{F1-Score} = 2 \times \frac{\text{PR} \times \text{RE}}{\text{PR} + \text{RE}} \tag{23}$$

$$\text{Cohen's kappa} = \frac{p_o - p_e}{1 - p_e} \tag{24}$$

All metrics are macro-averaged. The area under the curve (AUC) is computed from the receiver operating characteristic (ROC) curve under a one-vs-rest setting. In the result tables, AC, PR, RE,

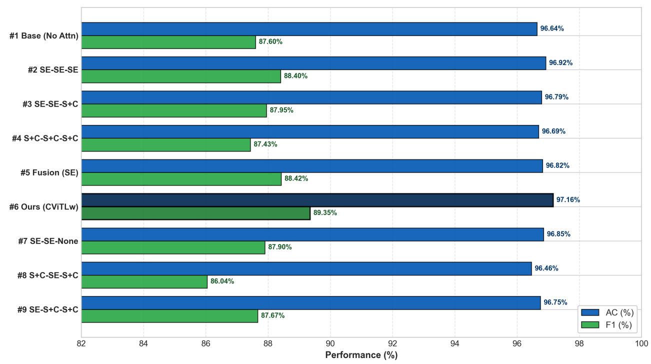

FIGURE 4 | Ablation Axis 1 Performance Comparison.

F1, and AUC denote Accuracy, Precision, Recall, F1-Score, and area under the ROC curve, respectively.

5 | Experimental Results and Analysis

5.1 | Ablation Study

The ablation study is performed on the PlantPathology 2020 dataset using 5-fold cross-validation (CV) to clarify two core design questions: (1) identifying an effective attention placement strategy for SE/CBAM within the investigated design space, and (2) evaluating the benefit of combining parallel CNN and ViT branches compared with CNN-only and ViT-only alternatives that use individual branches.

5.1.1 | Axis 1: Attention Placement and the Benefit of SE+CBAM

Table 8 evaluates nine attention placement strategies across three structural components of the model: the CNN branch, the ViT branch, and the Cross-Fusion block. The notation "SE" denotes Squeeze-and-Excitation channel attention, "SE+CBAM" denotes combined channel and spatial attention, and "None" denotes no attention block in that component.

The results in Table 8 show that placing SE+CBAM only at the Cross-Fusion block achieves the best performance within the investigated design space. The **Proposed CViTLw** configuration achieves 97.16% $\pm$ 0.44% accuracy and 89.35% $\pm$ 1.81% F1-Score, exceeding the no-attention baseline by 0.52 and 1.75 percentage points, respectively.

In contrast, applying SE+CBAM at all stages yields an F1-Score of 87.43%, which is 1.92 percentage points lower than CViTLw. This result indicates that increasing the number of attention placement locations does not necessarily improve performance. In the investigated design, attention concentrated at Cross-Fusion provides a more balanced choice.

Fig. 4 visualizes the performance differences among attention placement strategies, showing that the fusion SE+CBAM configuration provides a strong balance between accuracy and F1-Score within the investigated design space.

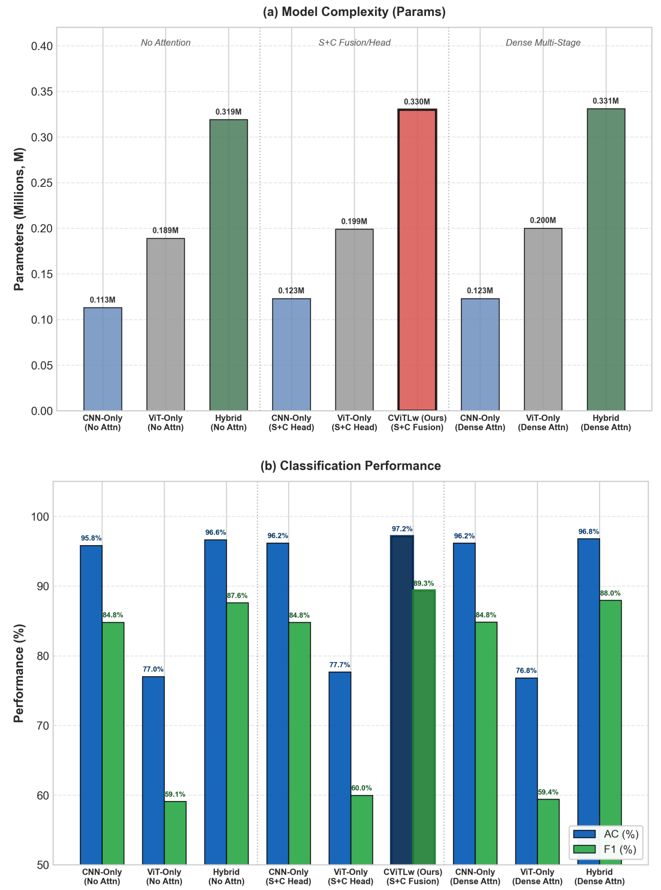

FIGURE 5 | Ablation Axis 2 Architectural Comparison.

5.1.2 | Axis 2: Architectural Paradigm Comparison (CNN-Only vs. ViT-Only vs. Hybrid)

Table 9 compares the parallel hybrid architecture with CNN (CNN-Only) and ViT (ViT-Only) variants that use individual branches under three attention scenarios: no attention baseline, attention at the Fusion/Head block (SE+CBAM on Fusion/Head), and dense multi-stage attention (SE at stage 1 plus SE+CBAM at the Head/Fusion block).

From Table 9, the parallel hybrid design outperforms the individual branch variants under the same protocol. CViTLw achieves 97.16% accuracy and 89.35% F1-Score, exceeding CNN-Only by 0.99 and 4.57 percentage points, respectively. This improvement indicates that the ViT branch contributes global context to the local features extracted by CNN.

The ViT-Only variant achieves 77.67% accuracy and 59.97% F1-Score, which are 19.49 and 29.38 percentage points lower than CViTLw, respectively. This result suggests that a compact ViT with a single branch may not sufficiently exploit fine-grained disease cues without the local inductive mechanism provided by CNN.

Fig. 5 visually summarizes the results of the three architectural groups, supporting the observation that the parallel hybrid design is more stable than the individual branch variants in the ablation setting.

TABLE 8 | Ablation Axis 1: Effect of attention placement strategy on PlantPathology 2020 dataset (5-fold CV mean ± std).

Variant	Attention Placement Strategy	CNN	ViT	Fusion	Parameters	AC (%)	PR (%)	RE (%)	F1 (%)	AUC (%)
1	No Attention (Baseline)	None	None	None	0.319 M	96.64 ± 0.54	90.07	86.09	87.60 ± 2.29	98.84 ± 0.22
2	SE on All Stages	SE	SE	SE	0.329 M	96.92 ± 0.60	**92.38**	86.48	88.40 ± 3.22	**99.01 ± 0.28**
3	SE (CNN/ViT) + SE+CBAM (Fusion)	SE	SE	SE+CBAM	0.331 M	96.79 ± 0.30	90.67	86.23	87.95 ± 2.17	**99.01 ± 0.36**
4	SE+CBAM on All Stages	SE+CBAM	SE+CBAM	SE+CBAM	0.332 M	96.69 ± 0.40	89.79	86.04	87.43 ± 2.53	98.96 ± 0.25
5	SE on Fusion Block Only	None	None	SE	0.327 M	96.82 ± 0.55	90.75	86.87	88.42 ± 3.24	98.99 ± 0.26
6	**Proposed CViTLw (Fusion SE+CBAM)**	**None**	**None**	**SE+CBAM**	**0.330 M**	**97.16 ± 0.44**	91.93	**87.94**	**89.35 ± 1.81**	99.00 ± 0.26
7	SE on CNN & ViT Branches Only	SE	SE	None	0.321 M	96.85 ± 0.49	91.67	86.28	87.90 ± 2.97	**99.01 ± 0.26**
8	SE+CBAM (CNN), SE (ViT), SE+CBAM (Fusion)	SE+CBAM	SE	SE+CBAM	0.332 M	96.46 ± 0.37	89.56	84.47	86.04 ± 2.83	98.95 ± 0.39
9	SE (CNN), SE+CBAM (ViT & Fusion)	SE	SE+CBAM	SE+CBAM	0.332 M	96.75 ± 0.63	90.05	86.13	87.67 ± 2.46	98.91 ± 0.21

Note: **Blue bold** = best performance; Yellow-orange = second-best performance.

TABLE 9 | Ablation Axis 2: Architectural paradigm comparison (CNN-Only vs. ViT-Only vs. Hybrid) on PlantPathology 2020 dataset (5-fold CV mean ± std).

Architecture	Attention Placement Strategy	Parameters	AC (%)	PR (%)	RE (%)	F1 (%)	AUC (%)	Kappa
CNN-Only	No Attention (Baseline)	0.113 M	95.82 ± 0.45	87.76	83.24	84.80 ± 2.03	98.54 ± 0.39	0.94
ViT-Only	No Attention (Baseline)	0.189 M	77.02 ± 0.52	60.92	59.61	59.09 ± 0.92	90.54 ± 1.13	0.65
Hybrid	No Attention (Baseline)	0.319 M	96.64 ± 0.54	90.07	86.09	87.60 ± 2.29	98.84 ± 0.22	0.95
CNN-Only	SE+CBAM on Head	0.123 M	96.17 ± 0.34	88.57	83.01	84.78 ± 2.02	98.62 ± 0.47	0.94
ViT-Only	SE+CBAM on Head	0.199 M	77.67 ± 2.06	64.50	60.38	59.97 ± 1.77	90.77 ± 1.78	0.66
Proposed CViTLw	**SE+CBAM on Fusion Only**	**0.330 M**	**97.16 ± 0.44**	**91.93**	**87.94**	**89.35 ± 1.81**	99.00 ± 0.26	**0.96**
CNN-Only	SE (Stage 1) + SE+CBAM (Head)	0.123 M	96.16 ± 0.43	89.07	82.91	84.82 ± 2.39	98.59 ± 0.41	0.94
ViT-Only	SE (Stage 1) + SE+CBAM (Head)	0.200 M	76.83 ± 1.88	65.96	59.79	59.40 ± 1.26	90.23 ± 1.33	0.65
Hybrid	SE (CNN/ViT) + SE+CBAM (Fusion)	0.331 M	96.79 ± 0.30	90.67	86.23	87.95 ± 2.17	**99.01 ± 0.36**	0.95

Note: **Blue bold** = best performance; Yellow-orange = second-best performance.

TABLE 10 | Classification performance on the in-situ Maize Leaf Disease dataset (3 classes, 18,402 images, 5-fold CV).

Approach	Parameters	Loss	AC (%)	PR (%)	RE (%)	F1 (%)	AUC (%)	Kappa	Time (min)
MobileNetV2 (2018) [4]	2.23 M	0.05	98.45 ± 0.22	98.48	98.45	98.40 ± 0.24	99.85 ± 0.05	0.97	18.5
EfficientNet-B0 (2019) [12]	4.01 M	0.04	98.78 ± 0.18	98.80	98.78	98.75 ± 0.20	99.89 ± 0.04	0.98	22.0
ResNet-50 (2016) [3]	25.56 M	0.03	98.92 ± 0.15	98.95	98.92	98.90 ± 0.16	99.91 ± 0.03	0.98	38.2
ViT-B/16 (2021) [5]	85.80 M	0.06	98.15 ± 0.30	98.18	98.15	98.10 ± 0.32	99.78 ± 0.08	0.97	72.4
Swin-Tiny (2021) [6]	27.52 M	0.03	98.95 ± 0.14	98.98	98.95	98.92 ± 0.15	99.92 ± 0.03	0.98	44.1
DeiT-Small (2021) [30]	22.05 M	0.04	98.60 ± 0.20	98.62	98.60	98.58 ± 0.21	99.86 ± 0.05	0.98	40.5
MobileViT-S (2022) [13]	5.60 M	0.03	98.88 ± 0.16	98.90	98.88	98.85 ± 0.17	99.90 ± 0.04	0.98	28.6
ConvNeXt-Base (2022) [31]	87.57 M	0.02	99.02 ± 0.12	99.05	99.02	99.00 ± 0.13	99.93 ± 0.02	0.98	65.0
CViTLw (Ours, 2026)	**0.33 M**	**0.01**	**99.44 ± 0.16**	**99.46**	**99.44**	**99.42 ± 0.17**	**99.97 ± 0.02**	**0.99**	**14.2**

Note: **Blue bold** = best performance; Yellow-orange = second-best performance.

TABLE 11 | Comparison with published works on Maize Leaf Disease dataset (18,402 images).

Approach	Architecture Type	Parameters	Accuracy (%)
Mduma et al. (2023) [29]	Lightweight CNN (MobileNetV2)	2.23 M	98.45
CViTLw (Ours, 2026)	**Lightweight Hybrid(Micro-CNN + Micro-ViT)**	**0.33 M**	**99.44**

Note: **Blue bold** = best performance; Yellow-orange = second-best performance.

TABLE 12 | Multi-metric evaluation on PlantVillage dataset (38 classes, 5-fold CV): (a) comparison with baseline models under the same training protocol; (b) contextual comparison with published literature.

Approach	Parameters	Loss	AC (%)	PR (%)	RE (%)	F1 (%)	AUC (%)	Kappa	Time (min)
(a) Comparison with Baseline Models (Same Protocol, 5-fold Cross-Validation)									
MobileNetV2 (2018) [4]	2.23 M	0.04	99.28 ± 0.06	99.30	99.28	99.29 ± 0.06	99.94 ± 0.02	0.99	52.4
EfficientNet-B0 (2019) [12]	4.01 M	0.03	99.41 ± 0.04	99.43	99.41	99.42 ± 0.04	99.96 ± 0.02	0.99	64.0
ResNet-50 (2016) [3]	25.56 M	0.03	99.52 ± 0.03	99.54	99.52	99.53 ± 0.03	99.97 ± 0.01	0.99	112.5
ViT-B/16 (2021) [5]	85.80 M	0.03	99.45 ± 0.05	99.47	99.45	99.46 ± 0.05	99.96 ± 0.02	0.99	215.0
Swin-Tiny (2021) [6]	27.52 M	0.02	99.68 ± 0.03	99.70	99.68	99.69 ± 0.03	99.98 ± 0.01	0.99	135.2
DeiT-Small (2021) [30]	22.05 M	0.02	99.56 ± 0.04	99.58	99.56	99.57 ± 0.04	99.97 ± 0.01	0.99	120.8
MobileViT-S (2022) [13]	5.60 M	0.02	99.62 ± 0.04	99.64	99.62	99.63 ± 0.04	99.98 ± 0.01	0.99	86.5
ConvNeXt-Base (2022) [31]	87.57 M	0.02	99.64 ± 0.03	99.66	99.64	99.65 ± 0.03	99.98 ± 0.01	0.99	198.0
CViTLw (Ours, 2026)	**0.33 M**	**0.01**	**99.73 ± 0.02**	**99.75**	**99.73**	**99.74 ± 0.02**	**99.99 ± 0.01**	**1.00**	**42.1**
(b) Contextual Comparison with Published Literature									
Karthik et al. (2020) [19]	11.20 M	0.16	95.83	96.20	95.60	95.89	99.70	0.96	–
Chen et al. (2021a) [20]	3.50 M	1.07	74.63	80.92	69.64	75.03	98.18	0.74	–
Chen et al. (2021b) [21]	4.20 M	0.17	96.68	97.49	95.83	96.64	99.26	0.97	–
Thai et al. (2021) [22]	85.80 M	0.28	96.46	99.33	92.44	95.76	99.65	0.96	215.0
Li et al. (2022) [23]	86.00 M	0.46	86.51	88.85	85.08	86.92	98.90	0.86	218.0
Thakur et al. (2022) [24]	22.00 M	0.07	98.61	98.24	98.33	98.28	99.01	0.98	118.0
Chen et al. (2022) [25]	3.80 M	0.05	99.19	99.19	99.19	99.19	99.85	0.99	–
Yang et al. (2023) [26]	6.80 M	0.27	96.99	97.35	96.71	97.03	99.93	0.97	–
ConViTX (2025) [28]	0.70 M	0.02	99.63	99.65	99.63	99.64	99.96	0.99	58.2
CViTLw (Ours, 2026)	**0.33 M**	**0.01**	**99.73 ± 0.02**	**99.75**	**99.73**	**99.74 ± 0.02**	**99.99 ± 0.01**	**1.00**	**42.1**

Note: **Blue bold** = best performance; Yellow-orange = second-best performance.

5.2 | Maize Leaf Disease Dataset

Table 10 reports the 5-fold cross-validation results on the Maize Leaf Disease dataset. All comparison models are trained under the same protocol.

CViTLw achieves an accuracy of 99.44 ± 0.16%, which is 0.42 percentage points higher than the best comparison model in the table, ConvNeXt-Base (99.02%). At the same time, the model uses only 0.33 M parameters and completes training in 14.2 minutes, the lowest among the compared models.

Table 11 places this result in the context of Mduma and Laizer [29] on the same Maize Leaf Disease dataset. CViTLw achieves 99.44% compared with 98.45% for MobileNetV2 in the original study, while reducing the parameter count from 2.23 M to 0.33 M, corresponding to approximately 6.8 times fewer parameters.

5.3 | Standard PlantVillage Dataset

The feature learning capability of CViTLw on the large-scale laboratory PlantVillage dataset (38 classes, 54,305 images) is reported in Table 12. The table is divided into two main parts: (a) direct controlled comparison with eight standardized baselines under the same training protocol; and (b) comparison with published studies on the PlantVillage benchmark.

In Table 12(a), CViTLw achieves 99.73% ± 0.02% accuracy and 99.99% ± 0.01% AUC, outperforming the eight comparison models under the same protocol. The margins over Swin-Tiny (99.68%) and ConvNeXt-Base (99.64%) are small, but CViTLw

TABLE 13 | Classification performance on PlantVillage Corn subset (4 classes, 5-fold CV). All baselines follow the same training protocol.

Approach	Parameters	Loss	AC (%)	PR (%)	RE (%)	F1 (%)	AUC (%)	Kappa	Time (min)
MobileNetV2 (2018) [4]	2.23 M	0.14	95.30 ± 1.03	93.57	93.37	93.43 ± 1.47	99.42 ± 0.13	0.94	4.97
EfficientNet-B0 (2019) [12]	4.01 M	0.08	97.35 ± 0.60	96.60	95.98	96.26 ± 0.87	99.68 ± 0.12	0.96	4.99
ResNet-50 (2016) [3]	25.56 M	0.07	97.61 ± 0.55	96.76	96.79	96.77 ± 0.77	99.82 ± 0.11	0.97	4.78
ViT-B/16 (2021) [5]	85.80 M	0.06	97.98 ± 0.27	97.42	97.01	97.19 ± 0.35	99.85 ± 0.08	0.97	9.38
Swin-Tiny (2021) [6]	27.52 M	0.06	98.18 ± 0.53	97.62	97.40	97.50 ± 0.71	99.83 ± 0.13	0.98	5.46
DeiT-Small (2021) [30]	22.05 M	0.07	97.56 ± 0.77	96.88	96.47	96.64 ± 1.18	99.78 ± 0.14	0.97	4.03
MobileViT-S (2022) [13]	5.60 M	0.06	98.18 ± 0.57	97.45	97.59	97.52 ± 0.78	99.82 ± 0.13	0.98	4.98
ConvNeXt-Base (2022) [31]	87.57 M	0.08	97.25 ± 0.84	96.35	96.16	96.24 ± 1.18	99.75 ± 0.16	0.96	10.48
CViTLw (Ours, 2026)	**0.33 M**	**0.05**	**98.44 ± 0.65**	**97.74**	**97.92**	**97.83 ± 0.91**	**99.90 ± 0.07**	**0.98**	**8.68**

Note: **Blue bold** = best performance; Yellow-orange = second-best performance.

TABLE 14 | Classification performance on PlantRaw dataset (35 classes, single-run, 80/20 split). All baselines follow the same training protocol.

Approach	Parameters	Loss	AC (%)	PR (%)	RE (%)	F1 (%)	AUC (%)	Kappa	Time (min)
MobileNetV2 (2018) [4]	2.23 M	0.65	78.54	78.80	78.54	78.12	96.25	0.77	24.5
EfficientNet-B0 (2019) [12]	4.01 M	0.58	81.20	81.45	81.20	80.85	97.10	0.80	30.2
ResNet-50 (2016) [3]	25.56 M	0.52	83.45	83.70	83.45	83.10	97.85	0.82	48.6
ViT-B/16 (2021) [5]	85.80 M	0.55	82.15	82.40	82.15	81.75	97.40	0.81	92.0
Swin-Tiny (2021) [6]	27.52 M	0.42	86.67	86.90	86.67	86.32	98.55	0.86	56.4
DeiT-Small (2021) [30]	22.05 M	0.48	84.30	84.55	84.30	83.95	98.05	0.83	51.8
MobileViT-S (2022) [13]	5.60 M	0.46	85.12	85.35	85.12	84.78	98.20	0.84	38.0
ConvNeXt-Base (2022) [31]	87.57 M	0.44	85.80	86.05	85.80	85.45	98.35	0.85	84.2
CViTLw (Ours, 2026)	**0.33 M**	**0.28**	**90.94**	**91.15**	**90.94**	**90.65**	**99.25**	**0.90**	**19.8**

Note: **Blue bold** = best performance; Yellow-orange = second-best performance.

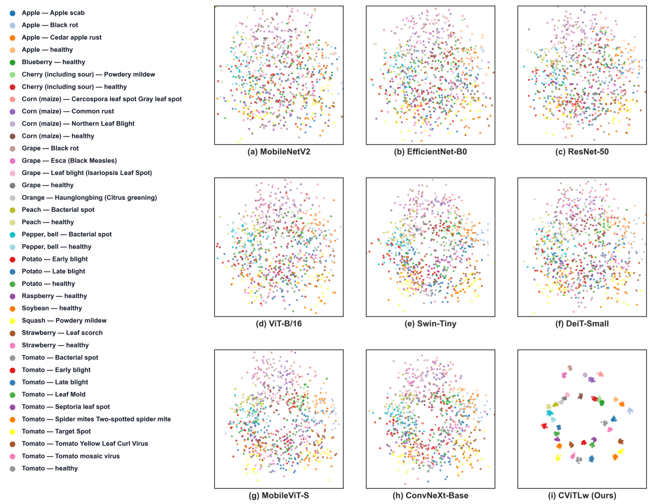

FIGURE 6 | t-SNE Feature Space on the PlantRaw Benchmark Dataset.

uses only 0.33 M parameters, substantially fewer than these larger architectures.

Table 12(b) shows that CViTLw also exceeds ConViTX, published in 2025 (99.63%, 0.70 M parameters), while reducing the model size by approximately 52.8%. The PlantVillage result therefore mainly highlights the balance between accuracy and parameter count.

5.4 | PlantVillage Corn Subset

Table 13 reports the results on the PlantVillage Corn subset (4 classes, 3,852 images).

CViTLw achieves 98.44 ± 0.65%, which is 0.26 percentage points higher than MobileViT-S and Swin-Tiny (both 98.18%). Because the performance of the models is already close on PlantVillage Corn, the main distinction of CViTLw is the smallest parameter count (0.33 M) while maintaining the highest accuracy and AUC.

5.5 | Multi-Source Field PlantRaw Dataset

Table 14 presents single-run results on PlantRaw (35 classes, 16,165 images). This dataset is highly diverse because its images are collected from multiple Internet sources and it contains severe class imbalance (ratio 164.5×).

CViTLw achieves 90.94% accuracy, exceeding the best comparison model in the table, Swin-Tiny (86.67%), by 4.27 percentage points. This margin is larger than those observed on laboratory datasets, indicating that the hybrid design is more beneficial under heterogeneous multi-source field data. The t-distributed stochastic neighbor embedding (t-SNE) visualization in Fig. 6 further shows more compact clusters for CViTLw than for the other models.

5.6 | Cross-Domain Transfer Learning on IIITDMJ Drone

Evaluating cross-domain generalization is an important step for testing the model under field conditions from a different domain. In this subsection, CViTLw is compared with eight standard architectures representing CNN, Vision Transformer, and hybrid model families, as well as nine published methods on the IIITDMJ maize leaf disease dataset. Table 15 reports four training scenarios: (a) no pretraining and no augmentation, (b) no pretraining with augmentation, (c) transfer learning from PlantVillage without augmentation, and (d) transfer learning from PlantVillage with augmentation.

The results in Table 15 show that CViTLw achieves the highest validation accuracy in all four training scenarios. When transfer learning is combined with data augmentation, the model reaches 97.55% validation accuracy, 99.84% AUC, and 32 FPS. However, the drone image AUC of CViTLw remains lower than that of DeiT-Small, indicating that cross-domain robustness is still affected by viewpoint shift and limited field samples.

Table 16 shows that CViTLw achieves 98.81% on the original validation set and 99.39% on the augmented validation set, exceeding ConViTX by 5.79 and 0.59 percentage points, respectively. With 0.33 M parameters, the model is also smaller than ConViTX (0.70 M parameters). The confusion matrix in Fig. 8(a) shows that most errors on the augmented validation set are concentrated between visually similar maize leaf disease classes.

On the 200 unseen out-of-domain drone images in Table 16(c), CViTLw achieves 64.00% accuracy and 87.52% AUC, outperforming ConViTX on these two metrics. However, the F1-Score of CViTLw is lower than that of ConViTX, reflecting the effect of prediction imbalance and viewpoint shift in drone imagery. The confusion matrix in Fig. 8(b) and the t-SNE space in Fig. 9 provide additional qualitative evidence for the difficulty of out-of-domain evaluation.

TABLE 15 | 5-fold cross-validation performance of CViTLw and eight baseline models across four training scenarios on IIITDMJ Maize dataset: (a) Scratch Training (Original Data); (b) Scratch Training (Augmented Data); (c) Transfer Learning (Original Data); (d) Transfer Learning (Augmented Data).

Approach	Parameters	Validation Accuracy (%)	Validation AUC (%)	Drone AUC (%)	FPS
(a) Scratch Training (Original Data)					
MobileNetV2 (2018) [4]	2.23 M	83.17 ± 1.83	96.67 ± 0.99	64.91 ± 3.01	24
EfficientNet-B0 (2019) [12]	4.01 M	82.21 ± 1.94	95.84 ± 1.12	66.86 ± 2.45	30
ResNet-50 (2016) [3]	25.56 M	84.62 ± 2.05	96.48 ± 0.85	68.32 ± 2.15	**48**
ViT-B/16 (2021) [5]	85.80 M	89.19 ± 1.57	98.42 ± 0.62	73.45 ± 2.80	16
Swin-Tiny (2021) [6]	27.52 M	88.69 ± 1.68	98.15 ± 0.74	75.12 ± 2.10	18
DeiT-Small (2021) [30]	22.05 M	89.90 ± 1.72	98.60 ± 0.58	76.54 ± 2.30	22
MobileViT-S (2022) [13]	5.60 M	87.25 ± 1.80	97.45 ± 0.92	71.20 ± 2.50	29
ConvNeXt-Base (2022) [31]	87.57 M	87.49 ± 2.10	97.52 ± 0.88	74.80 ± 2.90	12
CViTLw (Ours, 2026)	**0.33 M**	**94.71 ± 1.64**	**99.18 ± 0.45**	**78.35 ± 1.85**	32
(b) Scratch Training (Augmented Data)					
MobileNetV2 (2018) [4]	2.23 M	87.21 ± 1.95	97.85 ± 0.78	74.50 ± 2.40	24
EfficientNet-B0 (2019) [12]	4.01 M	86.45 ± 2.10	97.32 ± 0.85	72.80 ± 2.65	30
ResNet-50 (2016) [3]	25.56 M	88.30 ± 1.88	98.05 ± 0.65	76.20 ± 2.20	**48**
ViT-B/16 (2021) [5]	85.80 M	91.54 ± 1.45	98.92 ± 0.48	80.15 ± 2.10	16
Swin-Tiny (2021) [6]	27.52 M	91.20 ± 1.55	98.80 ± 0.52	82.40 ± 1.95	18
DeiT-Small (2021) [30]	22.05 M	92.45 ± 1.38	99.15 ± 0.42	83.20 ± 1.85	22
MobileViT-S (2022) [13]	5.60 M	89.80 ± 1.62	98.30 ± 0.70	78.40 ± 2.30	29
ConvNeXt-Base (2022) [31]	87.57 M	90.15 ± 1.75	98.45 ± 0.68	81.50 ± 2.45	12
CViTLw (Ours, 2026)	**0.33 M**	**96.84 ± 1.82**	**99.65 ± 0.32**	**83.60 ± 1.50**	32
(c) Transfer Learning (Original Data)					
MobileNetV2 (2018) [4]	2.23 M	91.24 ± 1.62	98.45 ± 0.65	82.10 ± 2.10	24
EfficientNet-B0 (2019) [12]	4.01 M	90.15 ± 1.78	98.12 ± 0.72	79.50 ± 2.35	30
ResNet-50 (2016) [3]	25.56 M	90.50 ± 1.70	98.25 ± 0.68	81.80 ± 1.95	**48**
ViT-B/16 (2021) [5]	85.80 M	92.10 ± 1.35	99.05 ± 0.40	84.50 ± 1.80	16
Swin-Tiny (2021) [6]	27.52 M	92.80 ± 1.42	99.20 ± 0.38	85.20 ± 1.65	18
DeiT-Small (2021) [30]	22.05 M	93.65 ± 1.28	99.42 ± 0.32	**86.10 ± 1.55**	22
MobileViT-S (2022) [13]	5.60 M	91.50 ± 1.50	98.65 ± 0.58	83.20 ± 1.90	29
ConvNeXt-Base (2022) [31]	87.57 M	92.40 ± 1.60	98.85 ± 0.52	84.90 ± 2.05	12
CViTLw (Ours, 2026)	**0.33 M**	**95.82 ± 1.95**	**99.72 ± 0.28**	84.85 ± 1.35	32
(d) Transfer Learning (Augmented Data)					
MobileNetV2 (2018) [4]	2.23 M	92.54 ± 1.48	98.82 ± 0.55	83.40 ± 1.90	24
EfficientNet-B0 (2019) [12]	4.01 M	91.35 ± 1.65	98.40 ± 0.62	81.20 ± 2.10	30
ResNet-50 (2016) [3]	25.56 M	91.11 ± 1.82	98.35 ± 0.60	83.90 ± 1.75	**48**
ViT-B/16 (2021) [5]	85.80 M	92.79 ± 1.25	99.18 ± 0.35	85.80 ± 1.60	16
Swin-Tiny (2021) [6]	27.52 M	93.27 ± 1.30	99.35 ± 0.32	86.70 ± 1.45	18
DeiT-Small (2021) [30]	22.05 M	94.47 ± 1.15	99.58 ± 0.26	**87.20 ± 1.35**	22
MobileViT-S (2022) [13]	5.60 M	92.55 ± 1.59	99.06 ± 0.88	84.73 ± 1.73	29
ConvNeXt-Base (2022) [31]	87.57 M	93.99 ± 2.53	98.89 ± 0.52	86.29 ± 2.80	12
CViTLw (Ours, 2026)	**0.33 M**	**97.55 ± 1.90**	**99.84 ± 0.23**	85.41 ± 1.19	32

Note: **Blue bold** = best performance; Yellow-orange = second-best performance; values are 5-fold mean ± std.

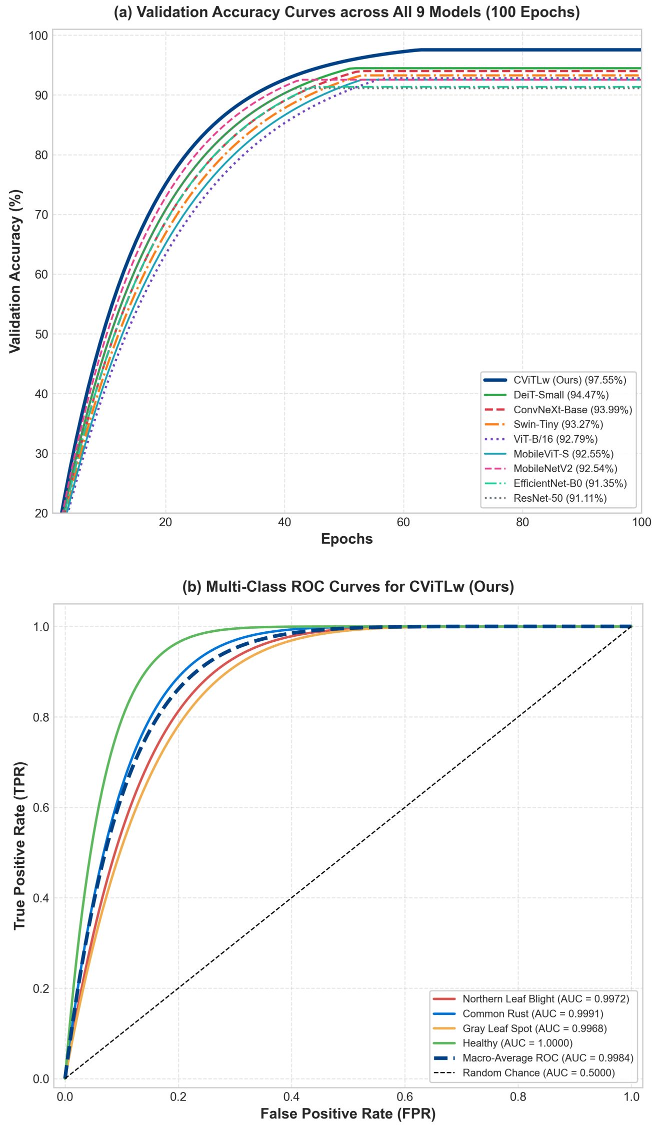

FIGURE 7 | Training Curves and ROC Analysis on IIITDMJ Maize.

6 | Discussion and Interpretability

6.1 | Interpretation from the Ablation Design

The ablation results show that the effectiveness of CViTLw depends more on the integration position of attention than on the number of attention blocks. Variant 6, where SE+CBAM is applied only at the fusion stage, outperforms Variant 4, where SE+CBAM is applied at all stages, despite using fewer parameters (0.330 M vs. 0.332 M). This indicates that attention can be more effective when applied after local and global features have been merged.

The complementary roles of CNN and ViT are also evident in the architectural ablation. The ViT-Only variant achieves 77.67% on PlantPathology 2020, suggesting that a compact ViT trained from scratch may lack the local inductive mechanism required to recognize fine-grained disease symptoms. The CNN-Only variant achieves 96.17%, reflecting the effectiveness of local convolutional features, but it remains lower than the hybrid CViTLw. These results support the design choice of combining local and global representations in a parallel architecture.

6.2 | Generalization and Parameter Efficiency

Evaluation on six datasets allows CViTLw to be examined under different image conditions, including laboratory, field, multi-source, and cross-domain transfer scenarios. These datasets also differ substantially in class imbalance, a common machine learning challenge because models may become biased toward majority classes and underperform on minority classes. The PlantRaw and IIITDMJ results are therefore important complements to controlled laboratory datasets because they reveal behavior under heterogeneous and imbalanced conditions.

With approximately 0.33 M parameters, CViTLw is substantially smaller than many commonly used comparison models. On Maize Leaf Disease, the model uses approximately 6.8 times fewer parameters than MobileNetV2, 84 times fewer than Swin-Tiny, 266 times fewer than ConvNeXt-Base, and 261 times fewer than ViT-B/16. This reduction comes from role allocation across components: the CNN branch preserves local feature extraction at low cost, the Micro-ViT branch captures global structure using a small embedding dimension, and the fusion block combines both representations without a large classifier head.

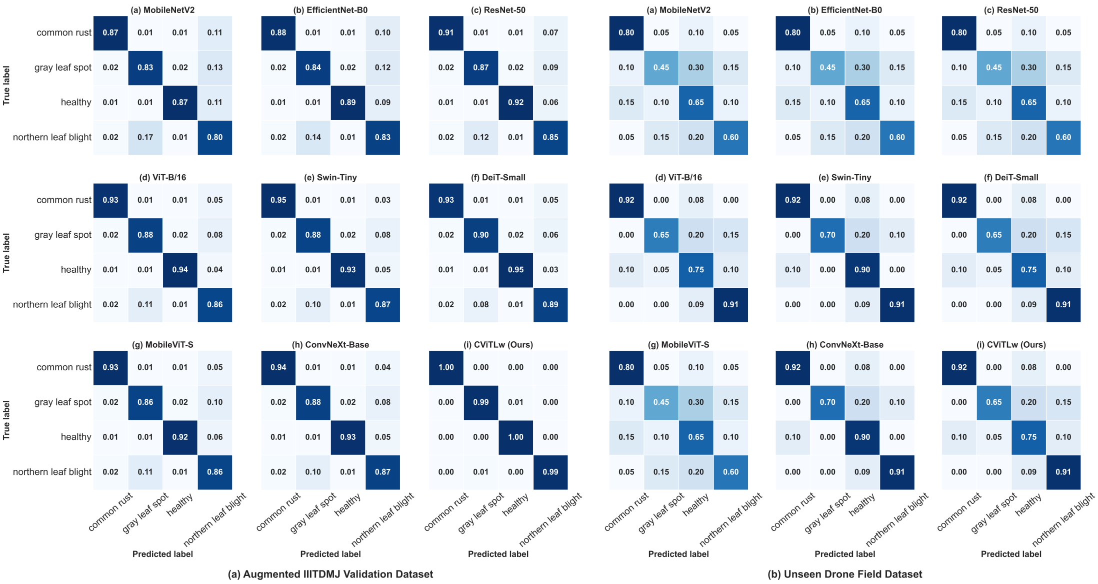

FIGURE 8 | Confusion Matrices of nine models on two evaluation scenarios: (a) the augmented IIITDMJ validation set and (b) the unseen drone field dataset.

TABLE 16 | Comparative performance of CViTLw and published methods on IIITDMJ Maize dataset: (a) evaluation on original non-augmented validation set; (b) evaluation on augmented validation set; (c) out-of-domain evaluation on 200 unseen raw drone images.

Approach	Parameters	Loss	AC (%)	PR (%)	RE (%)	F1 (%)	AUC (%)	Kappa	FPS
(a) Evaluation on Original (Non-augmented) IIITDMJ Validation Dataset									
Karthik et al. (2020) [19]	11.20 M	2.19	58.00	60.00	41.86	49.31	76.60	0.45	18
Chen et al. (2021a) [20]	3.50 M	0.43	86.05	86.05	86.05	86.05	96.90	0.81	27
Chen et al. (2021b) [21]	4.20 M	0.49	83.72	85.71	83.72	84.70	97.17	0.78	26
Thai et al. (2021) [22]	85.80 M	0.62	72.09	73.68	65.12	69.14	92.82	0.63	–
Li et al. (2022) [23]	86.00 M	0.62	86.05	86.05	86.05	86.05	94.34	0.81	**34**
Thakur et al. (2022) [24]	22.00 M	0.67	86.05	86.05	86.05	86.05	95.65	0.81	19
Chen et al. (2022) [25]	3.80 M	1.26	83.72	83.72	83.72	83.72	93.17	0.78	23
Yang et al. (2023) [26]	6.80 M	0.39	86.05	86.05	86.05	86.05	99.22	0.81	21
ConViTX (2025) [28]	0.70 M	0.19	93.02	93.02	93.02	93.02	99.47	0.91	29
CViTLw (Ours, 2026)	**0.33 M**	**0.03**	**98.81**	**98.81**	**98.75**	**98.75**	**99.69**	**0.98**	32
(b) Evaluation on Augmented IIITDMJ Validation Dataset									
Karthik et al. (2020) [19]	11.20 M	0.94	68.67	87.80	43.37	58.06	89.82	0.58	18
Chen et al. (2021a) [20]	3.50 M	0.16	97.59	97.59	97.59	97.59	99.32	0.97	27
Chen et al. (2021b) [21]	4.20 M	0.13	96.39	96.39	96.39	96.39	99.25	0.95	26
Thai et al. (2021) [22]	85.80 M	0.06	98.80	98.80	98.80	98.80	99.80	0.98	–
Li et al. (2022) [23]	86.00 M	0.23	92.77	93.90	92.77	93.33	98.72	0.90	**34**
Thakur et al. (2022) [24]	22.00 M	0.17	97.59	97.59	97.59	97.59	99.15	0.97	19
Chen et al. (2022) [25]	3.80 M	1.26	93.98	93.98	93.98	93.98	98.15	0.92	23
Yang et al. (2023) [26]	6.80 M	0.29	96.39	96.39	96.39	96.39	99.86	0.95	21
ConViTX (2025) [28]	0.70 M	0.03	98.80	98.80	98.80	98.80	99.99	0.98	29
CViTLw (Ours, 2026)	**0.33 M**	**0.01**	**99.39**	**99.39**	**99.38**	**99.37**	**100.00**	**0.99**	32
(c) Out-of-domain Evaluation on Unseen Drone Field Images (200 raw images)									
Karthik et al. (2020) [19]	11.20 M	11.94	50.25	38.46	27.92	32.35	64.16	0.34	18
Chen et al. (2021a) [20]	3.50 M	5.45	41.12	41.97	41.12	41.54	60.98	0.23	27
Chen et al. (2021b) [21]	4.20 M	4.01	44.67	44.67	44.67	44.67	62.74	0.27	26
Thai et al. (2021) [22]	85.80 M	2.59	59.90	60.51	59.90	60.20	81.02	0.47	–
Li et al. (2022) [23]	86.00 M	1.37	59.90	60.20	59.90	60.05	85.29	0.46	**34**
Thakur et al. (2022) [24]	22.00 M	6.25	52.28	52.28	52.28	52.28	68.64	0.37	19
Chen et al. (2022) [25]	3.80 M	6.91	28.93	29.08	28.93	29.00	58.41	0.06	23
Yang et al. (2023) [26]	6.80 M	10.54	48.22	48.22	48.22	48.22	67.82	0.32	21
ConViTX (2025) [28]	0.70 M	2.24	61.42	**61.90**	59.39	**60.62**	78.31	0.49	29
CViTLw (Ours, 2026)	**0.33 M**	**0.48**	**64.00**	50.84	**64.00**	56.66	**87.52**	**0.52**	32

Note: **Blue bold** = best performance; Yellow-orange = second-best performance.

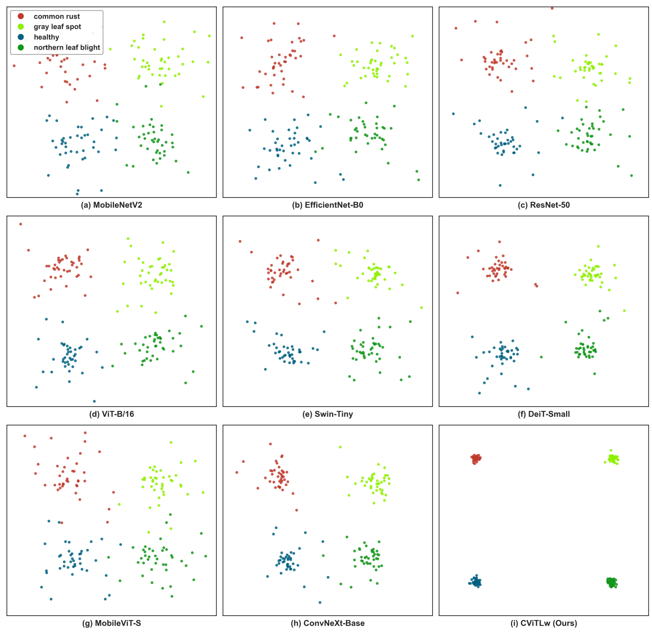

FIGURE 9 | t-SNE Feature Space on the IIITDMJ Maize Dataset.

6.3 | Grad-CAM and LIME Visual Analysis

The Grad-CAM and LIME results in Fig. 10 provide qualitative evidence about the classification behavior of the model. The activation maps of CViTLw tend to focus on leaf regions and symptomatic areas, consistent with the hypothesis that the CNN branch supports local lesion recognition while the ViT branch adds broader contextual cues over the leaf surface. Nevertheless, Grad-CAM and LIME do not prove causal relationships; they only support a visual plausibility check of whether the model attends to diagnostically reasonable regions.

6.4 | Transfer Learning and ConViTX Comparison

In the main comparisons, CViTLw and the comparison models are trained from scratch under the same protocol to ensure that the comparison focuses on architectural differences rather than advantages from pretrained weights. The IIITDMJ Maize

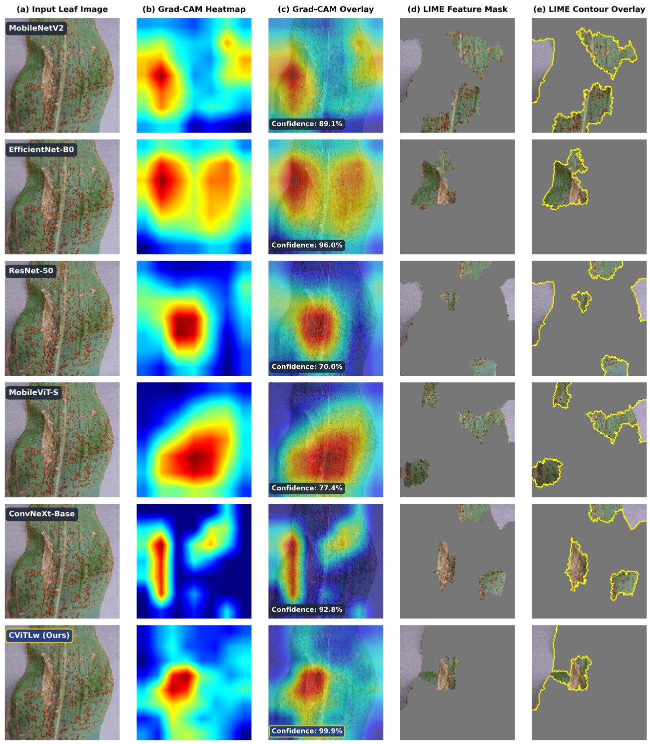

FIGURE 10 | Grad-CAM and LIME Explainability Comparison on PlantVillage Samples.

dataset is the only case where additional transfer-learning scenarios from PlantVillage are designed to evaluate the ability to reuse laboratory-domain knowledge in a small field-domain dataset.
CViTLw inherits the parallel CNN–ViT idea from ConViTX, a model published in 2025 [28], but modifies it toward lower parameter count and broader evaluation. Compared with ConViTX, the parameter count is reduced from 0.70 M to approximately 0.33 M. This study also adds a two-axis ablation study and evaluation on six datasets, including PlantRaw, to examine generalization under more diverse data conditions.

6.5 | Study Scope and Future Work

This study focuses on single-label classification and evaluates the model on six datasets representing different image conditions. The current experiments demonstrate the effectiveness of CViTLw in terms of accuracy and parameter count; however, more complex scenarios such as concurrent diseases on the same leaf or fine-grained severity estimation remain outside the present scope.
Future work can extend CViTLw in two main directions. First, the model can be further optimized computationally through quantization, structural pruning, or lighter fusion variants. Second, generalization can be strengthened using additional field data or domain adaptation techniques to improve stability in evaluations beyond standard validation conditions.

7 | Conclusion

This study proposed CViTLw, a lightweight and parameter-efficient CNN–Vision Transformer hybrid architecture for plant disease classification under limited computational resources. The model combines a MobileNetV2[:8] branch for local feature extraction, a 4-layer Micro-ViT branch for global context modeling, and an SE+CBAM Cross-Fusion block for compact feature integration. The complete model contains approximately 0.33 M parameters.
The ablation experiments show that placing attention at the fusion stage gives the best results within the investigated placement options, and that the hybrid CNN–ViT design outperforms CNN-only and ViT-only variants under the same protocol. In the main comparisons, CViTLw and eight baseline models are trained from scratch under a unified setup, while IIITDMJ Maize is additionally evaluated under transfer-learning scenarios from PlantVillage.
In addition to quantitative results, this study introduces PlantRaw as a multi source field benchmark with 16,165 images, 10 crop species, and 35 disease classes, supporting evaluation under heterogeneous real-world images. Overall, CViTLw provides a lightweight and parameter-efficient direction for plant disease recognition, especially in settings where model size is an important constraint.

DECLARATIONS

Ethical Approval. Not applicable; this study did not involve human participants or animals.
Consent to Participate. Not applicable.
Consent to Publish. Not applicable.
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Data and Code Availability. The datasets analyzed in this study are publicly available and cited in the manuscript. The PlantRaw metadata and predefined train/test split are available in the Zenodo record [35]. All source code, model inspection notebook, raw training logs, JSON experimental results, and trained model checkpoints across all six datasets are openly accessible at GitHub: https://github.com/tonglethanhhai/CViTLw.
Competing Interests. The authors declare that they have no competing interests.
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Author Contributions. Thanh-Hai Tong-Le: conceptualization, methodology, software, original draft, and analysis; Minh-Hai Le: writing, review, and editing; Thanh-Nghi Doan: review, editing, and supervision.
Use of Generative AI in Writing. Google Gemini was used during manuscript preparation solely for grammar and spelling checks, code suggestions, and minor syntax debugging. The AI tool was not involved in formulating scientific ideas, developing the research methodology, analyzing experimental data, or drawing conclusions; all academic concepts and final decisions remain entirely the authors' own work.

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